LINING YAN

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Research Interests: Robotic Learning · Multimodal Retrieval · Efficient Deep Learning · Agentic Workflows · NLP

EDUCATION

Yale University — M.S. in Computer Science *New Haven, CT | Sep 2026 – Expected 2027*
Graduate study focused on AI, robotics, and machine learning; preparing for research in embodied and interactive intelligent systems.
Selected Coursework: Intelligent Robotics, Advanced Robotic Mechanisms, Sequential Decision Making, Trustworthy Deep Learning, Advances in Frontier AI Models.

Kean University — B.S. in Computer Science, Minor in Applied Mathematics *Union, NJ, USA | 2022 – 2026*
GPA: 3.92/4.0 · Summa Cum Laude · Dean's List (2023, 2024, 2025)
Selected Coursework: Algorithms & Data Structures, Advanced Machine Learning, Generative AI, Big Data Analytics, Numerical Analysis.

Stanford University — Summer Session *Stanford, CA, USA | Jun – Aug 2024*
GPA: 4.12/4.3. Coursework: Introduction to High-Performance Computing, HPC-AI Summer Seminar Series, Data Mining and Analysis, Introduction to Stochastic Processes I.

SKILLS

Programming: Python, Java, SQL, NumPy, pandas, scikit-learn
AI & Machine Learning: PyTorch, Transformers, Deep Learning, NLP, Multimodal Learning
Tools & Computing: Git, Linux, Docker, HPC
Languages: English (Fluent), Mandarin Chinese (Native)

RELEVENT EXPERIENCE

Wenzhou-Kean University — NLP Research Group, Undergraduate Researcher (Advisor: Dr. Baha Ihnaini) *Zhejiang, China | 2023 – 2025*

- Conducted two years of advised research on AI-generated text detection and interpretability, spanning global/local linguistic feature analysis, ensemble modeling, and semi-supervised training under limited labels.
- *Primary research output:* Multi-Level Text Feature Analysis for Detecting AI-Generated Content and Semi-Supervised Artificial Intelligence Text Detection with RoBERTa.

University of Notre Dame — Center for Research Computing, Research Intern (Advisor: Dr. Samuel Rund) *Notre Dame, IN | Jul – Aug 2026*

- Built an agentic pipeline converting scientific PDFs into structured evidence (OCR, layout/figure-table detection, Ask QA) targeting biology/epidemiology traits such as study design, sample size, population, intervention, and outcome.
- Designed multi-step extraction workflows over text, tables, and figures with reliability gates (metric/life-stage binding, multi-part completeness, Methods-vs-Results retrieval, false "Not stated" repair) to reduce hallucinated or cross-experiment answers.
- Implemented explore-vs-full processing modes and on-demand Ask extraction, allowing large corpora to be scanned quickly and deepened only when users need grounded tables, charts, or answers.

National Research Institute — Research Intern (Advisor: Dr. Guoxin Kang) *Beijing, China | Jan – Jun 2026*

- Researched multimodal generative retrieval, investigating alternatives to autoregressive identifier generation for large-scale retrieval systems.
- Formulated semantic bucket assignment as an entropy-regularized optimal transport problem; implemented Sinkhorn-based optimization to enforce balanced routing.
- Designed and evaluated efficient retrieval architectures combining learned routing, local candidate filtering, and exact similarity reranking.

PUBLICATIONS & MANUSCRIPTS

Pattern-Centric Detection of AI-Generated Text: An Interpretable and Data-Efficient Expert System *In preparation*

- **Author (Senior Capstone).** Built a pattern-centric expert system — a KMeans-derived Generative Pattern Library over multi-granularity linguistic features paired with an Explainable Boosting Machine (EBM) — achieving AUROC above 0.99 using only 44–50 labeled samples across five benchmarks (DAIGT, HC3, LLMDetect, MAGE, MultiModel), retaining over 95% of full-model performance at ~1/100th the labeling cost and outperforming zero-shot DetectGPT on all five.
- Discriminative across 27+ LLM families and adversarial perturbations without white-box model access (applicable to closed-source LLMs); every prediction carries an auditable SHAP feature attribution, and a modular pattern library enables rapid domain adaptation without full pipeline retraining.

Multi-Level Text Feature Analysis for Detecting AI-Generated Content *CSCI 2025*

- **First Author.** Analyzed AI-generated vs. human-written abstracts using global and local text features; developed a fine-tuned BERT + Random Forest ensemble achieving 99.9% (abstract-level) and 93.5% (sentence-level) F1. Used SHAP analysis to identify perplexity and log-probability as key discriminative features.

Bridging the Modality Gap that Hurts Cross-Modal Retrieval *Submitted, VLDB 2026*

- **Co-First Author.** Proposed a non-autoregressive multimodal retrieval framework using Sinkhorn-balanced semantic buckets, a 2-layer MLP router, bucket-local CE filtering, and exact cosine reranking. Achieved R@1 = 53.6% (MS-COCO) and R@5 = 46.7% (Visual News) while exploring ~1.4% of the corpus; router latency of 0.15 ms/query, 13× faster end-to-end inference than GENIUS on Visual News.

MCP-X: An Ultra-Compact CNN for Rice Disease Classification *AgriEngineering 2025, 7(11), 359*

- **Second Author.** Designed a 0.21M-parameter CNN for real-time rice disease classification in resource-constrained settings; achieved 98.93% validation accuracy on PlantVillage, outperforming MobileNetV2 and Inception-V3 with far fewer parameters.

Semi-Supervised Artificial Intelligence Text Detection with RoBERTa *In preparation*

- **Lead Researcher.** Developed a semi-supervised RoBERTa framework achieving up to 97.79% F1 on the HC3 dataset using only 1% labeled data, outperforming BERT, ELECTRA, ModernBERT, and baseline RoBERTa.

AI-Enhanced Echocardiography: Comparative Analysis of Image Quality Using ESRGAN *In preparation*

- **Lead Researcher.** Applied generative AI to improve echocardiographic image quality and clinical-metric accuracy across Original, CycleGAN, ESRGAN, and a novel ESRGAN-Denoise pipeline; ESRGAN-Denoise achieved the best segmentation (Dice 0.9670, IoU 0.9362) and clinical alignment, indicating strong potential for clinical deployment.

COURSE PROJECTS

CPU Scheduling Algorithms Simulator — Operating Systems *Fall 2024*

- Developed a GUI-based simulator to analyze and compare CPU scheduling algorithms (FCFS, SJF, Priority Scheduling, Round Robin, Multilevel Queue Scheduling) as an interactive educational tool for process management concepts.

Romberg Integration System with Improper Integral Handling — Numerical Analysis *Fall 2024*

- Built a Python-based system computing integrals via the Romberg method, with enhancements to handle improper integrals and singularities across a wide range of functions.

TEACHING & CAMPUS SERVICE

Wenzhou-Kean University — Peer Tutor *Zhejiang, China | Oct 2023 – Dec 2025*

- Provided one-on-one guidance on learning methods and course material to 80+ students; met regularly with academic support center staff to track student progress.

Wenzhou-Kean University — Computer Laboratory Assistant *Zhejiang, China | Oct 2023 – Dec 2025*

- Checked and maintained laboratory equipment; conducted weekly equipment test records and reports.

HONORS & AWARDS

- Best Poster Award, Student Research Day (2024)
- Scholarship for Research and Innovation (2024)
- Third Prize, AI Shark Tank Contest — AI & University Life (2024)
- Dean's Second Prize Scholarship (2023, 2024, 2025)
- Zhejiang Provincial Government Scholarship (2024, 2025)
- Honorable Mention, US Collegiate Mathematical Contest in Modeling (2024)
- Outstanding Graduate of Zhejiang Province (2026)